Lecture 06 — Linear functions in applications

Fri Sep 4

Last updated: September 2, 2026.

Reading: Boyd 2.1–2.3 applications (score, cost, temperature, regression). Quiz 2 prep: Boyd 1.12, 1.14, 1.15. Related objectives: Lectures. Schedule. Quiz 2 is Friday, September 11.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Check yourself**.

Learning objectives

By the end of class, students should be able to:

1. Model a score, cost, prediction, or aggregate quantity.
2. Interpret model coefficients and intercepts.
3. Compare outputs from competing models.
4. State a reasonable limitation or assumption of a linear model.

Warm-up

Is $f(x) = 5x_1 + 2x_2$ linear, affine, or both? What about $g(x) = 5x_1 + 2x_2 - 3$?

Fill in

An **affine model** $f(x) = a^\mathsf{T} x + b$ often approximates a real quantity only over a limited range of inputs; outside that range the model may be ______________.

In a cost model $c^\mathsf{T} q$, the entry c_i is the cost per unit of material i. The total cost is a ______________ of the amounts q_i.

In Boyd 1.14, $f^\mathsf{T} h$ is **sector exposure**: net dollars invested in assets with $f_i = 1$. If $f^\mathsf{T} h = 0$, the portfolio is ______________ to that sector.

To pick the cheapest of K suppliers with price vectors $p_1, \dots, p_K$ for order q, compare $p_1^\mathsf{T} q, \dots, p_K^\mathsf{T} q$ and choose the ____________.

A regression model $\hat{y} = \beta^\mathsf{T} x + v$ is useful when the relationship is roughly ______________ over the data range, but coefficients can be hard to interpret if features are ______________.

Worked in class

W1. A 30-question magazine quiz uses answers $a_i \in \{1, 2, 3\}$ (Rarely, Sometimes, Often). Questions 1–15: Sometimes +1, Often +2. Questions 16–30: Sometimes +2, Often +4. Rarely adds 0. Write total score $s = w^\mathsf{T} a + v$ (Boyd 2.6).

W2. Cash held in five currencies is $c = (1000, 500, -200, 0, 300)$ (USD, RMB, EUR, GBP, JPY). Exchange rates to USD are $r = (1, 0.14, 1.08, 1.27, 0.0067)$. Total USD value is $r^\mathsf{T} c$ (Boyd 1.12). Compute it (negative EUR is a liability).

W3. Three assets; pharma indicator $f = (1, 0, 1)$. Portfolio $h = (500, 200, -100)$ dollars (short on asset 2). Compute exposure $f^\mathsf{T} h$ (Boyd 1.14). In words, what does the sign mean?

W4. Order $q = (10, 5, 20)$ of three materials. Supplier A prices $p_A = (2, 4, 1)$; supplier B prices $p_B = (3, 2, 2)$ per unit (Boyd 1.15). Which supplier is cheaper for the whole order? Compare $p_A^\mathsf{T} q$ and $p_B^\mathsf{T} q$.

W5. Processor temperature is affine in power: $T = a^\top P + b$. Idle $P = (10, 10, 10)$ gives $T = 30$. Full power on processor 1 only, $P = (100, 10, 10)$, gives $T = 60$. Find a_1 from those two readings (Boyd 2.2 setup).

Check yourself

C1. Two regression models for the same house: $\hat{y}_A = 200{,}000 + 120x_1 + 15{,}000x_2$ and $\hat{y}_B = 180{,}000 + 130x_1 + 18{,}000x_2$, where x_1 is area (1000 ft^2) and x_2 is bedrooms. For $x = (2, 3)$, compute both predictions. Which predicts higher?

C2. Boyd 2.4 style: $\varphi(1, 1, 0) = -1$, $\varphi(-1, 1, 1) = 1$, $\varphi(1, -1, -1) = 1$. Could φ be linear? Could it be affine? Brief reason.

C3. You must buy $q = (4, 2)$. Supplier 1: $p_1 = (5, 3)$; supplier 2: $p_2 = (4, 5)$. Which supplier wins on total cost $p_k^\top q$?

C4. A long-only portfolio ($h_i \geq 0$) is neutral to pharma ($f^\mathsf{T} h = 0$ with $f_i \in \{0, 1\}$). In one sentence, what does that mean about pharma holdings?

C5. Name one reasonable **limitation** of using a linear model for café revenue $p^\mathsf{T} x$ when x counts drinks sold.

Answers (Check yourself)

C1. $\hat{y}_A = 200{,}000 + 120 \cdot 2 + 15{,}000 \cdot 3 = 245{,}240$. $\hat{y}_B = 180{,}000 + 130 \cdot 2 + 18{,}000 \cdot 3 = 248{,}260$. Model B predicts higher.

C2. **Cannot be linear:** if $\varphi(x) = a^\mathsf{T} x$, the three given values force an inconsistent system for a. **Could be affine:** many choices of a, b with $\varphi(x) = a^\mathsf{T} x + b$ fit three input–output pairs.

C3. $p_1^\mathsf{T} q = 5 \cdot 4 + 3 \cdot 2 = 26$. $p_2^\mathsf{T} q = 4 \cdot 4 + 5 \cdot 2 = 26$. Tie.

C4. No net dollar holdings in pharma stocks (long positions in pharma are fully offset by non-pharma or the portfolio holds none).

C5. Sample: prices fixed but demand may saturate; loss leaders; does not model quantity discounts; assumes all drinks sold at listed price.